

PAVAN RAPOLU

+1 (940) 758-3931 • rapolu.pavann@gmail.com • [LinkedIn](#) • [Portfolio](#) • [GitHub](#) • Dallas, TX

EDUCATION

Master of Science in Computer Science, **University of North Texas**, Denton, Texas August 2021 – May 2023
Bachelor of Technology in Electronics and Communication Engineering, **MGIT**, Hyderabad, India August 2016 – June 2020

TECHNICAL SKILLS

Programming Languages: Python, R, Java, C++, GraphQL, SQL

Web Technologies: HTML5, CSS3, Bootstrap, Ag-grid, JavaScript, Angular 13, React, jQuery, XML, AJAX, Dash

Database Connectivity and Management: Oracle, MS Access, MYSQL Workbench, NoSQL, MongoDB, Dynamo DB, Cassandra

Backend and Cloud: Django, Flask, Django Rest, Hibernate, FastAPI, NodeJS, AZURE, AWS, Spark

Operating Systems & Others: GIT, JIRA, Eclipse, VS Code, NetBeans IDE, Pycharm, CI/CD, Splunk, Docker, Linux, Tableau, Figma

EXPERIENCE

TechAlpha LLC, *Python Developer*, Dallas, TX.

August 2023 – Present

- Engineered **Python** application with **Django**, achieving **20%** performance boost for real-time circuit parameter tracking.
- Led full-stack development using **React**, **HTML5**, **CSS3** enhancing dashboards for a **15%** increase in user engagement.
- Implemented **Django Migrations**, ensuring seamless updates and reducing database-related issues by **25%**.
- Optimized deployment with **Docker**, reducing time by **30%** and ensuring consistent performance across environments.
- Utilized Python modules (**Numpy**, **SciPy**, **Pandas**, **Pygal**) for graphical data representation, offering clear insights for data analysis. Integrated PEP coding conventions, improving code maintainability by **30%**.

University of North Texas, *Graduate Student Assistant*, Denton, TX.

December 2021 – May 2023

- Headed integration and optimization of web apps, achieving a **70%** latency reduction using **Django**, **XML**, **JS**, **PostgreSQL**.
- Applied design principles and modern **DB** architecture for **40%** **UI/UX** enhancement in landing pages deployment.
- Mitigated **GraphQL** query bottlenecks, realizing **30%** faster execution and ensuring consistent app efficiency.
- Enacted **CI/CD** pipelines via **CircleCI**, achieving a **50%** improvement in resource allocation.

Virtusa, *Python Full-Stack Developer*, Hyderabad, India.

July 2020 – August 2021

- Successfully addressed **100%** of server module issues through strategic design enhancements.
- Architected robust back-end solutions using **Python**, **flask**, and front-end technologies, resulting in a **30%** performance boost. Carried out seamless database management with PostgreSQL.
- Drove **40%** efficiency surge through **ReactJS** integration; formulated **Restful microservices** with **Django** on **AWS**.
- Demonstrated exemplary expertise in designing, implementing, and optimizing **GraphQL APIs** for web applications, contributing to a **20%** improvement in overall system efficiency.

Tech Mahindra, *Python Developer Intern*, Hyderabad, India.

May 2019 – April 2020

- Executed and maintained **GraphQL** schemas and resolvers, resulting in a notable **15%** reduction in data retrieval times.
- Collaborated with front-end developers to seamlessly integrate **GraphQL** into client applications, enhancing overall responsiveness by **25%**.
- Initiated the development of a **Django** application featuring a **RESTful API**, achieving a substantial 30% increase in data collection efficiency while ensuring impeccable data integrity.
- Conducted client-side validation using **JavaScript** and **jQuery**, ensuring **95%** data accuracy and integrity. Established a robust **unit test** framework, enhancing code reliability by **50%**.

ACADEMIC PROJECTS

Cloud based Face Recognition Application (Skills: Python, Flask, EC2, S3, DynamoDB, Raspberry Pi, OpenCV, TensorFlow)

- Accomplished data scientist specializing in NLP and sentiment analysis. Developed a 92% accurate Sentiment Analysis tool for diverse social media content and boosted data pipelines, reducing processing time by 30% [[code link](#)].

Smart Dustbin (Skills: C++, React, PyCharts, Django, Arduino, Python, GitHub)

- Developed a Robotic Smart Dustbin with 92% precision rate, leveraging robotics and data analysis expertise; revolutionized waste management efficiency and reduced manual labor by 40%. Implemented version control using GIT and deployed the system on AWS, achieving 30% scalability and reliability boost; enhanced waste management efficiency for streamlined operations and increased customer satisfaction. [[code link](#)].

ACHIEVEMENTS/ CERTIFICATIONS

- Received a competitive scholarship from the University of North Texas.
- Microsoft Certified: Azure Fundamentals, Certification Number: I783-5048, June 2023